Equations as images

> We drew contexts of the in-context grammar with pictures for operands°, so the model answers a whole image one pixel at a time. The unedited model paints a recognizable picture under every op, and the full edit° turns the `difference` picture into a speckle of the other ops while leaving the others alone. Walking the examples in step with the query, as a compositor would, gathers the stray pixels into patches that follow the example images.

Operands. Photos from the Kodak Lossless True Color suite, which Kodak released for unrestricted use, and two synthetic images…

Full edit. The e_1 component removed from the state at every position and every slice, as in the D2.2 design.

This is a figure study for the D2.2 post, with no new training. It reads the eight `τ_ℓ fixed` runs of the fixed-confirm° experiment and asks each of them about 1,376,256 contexts, one per pixel of each query image, op and way of putting the operands on the grid, with and without the edit.

fixed-confirm. Fixed confirmation…

Scope and method

Each pixel of a query image pair gets a context of its own. Its three examples are pixels from three example image pairs, each shown with its answer under the hidden op, and its query is the pixel pair itself. The model answers at the query `=`, and the answers over every pixel make a picture of the op as the model infers it. The examples are the same draws for every run and edit, so pictures from different runs and edits match pixel for pixel.

Operands

Photos from the Kodak Lossless True Color suite, which Kodak released for unrestricted use, and two synthetic images: a gray ramp from black to white, and a map of hue across and lightness down. Each is resized to 192 × 128 and put on the grid of 216 colors in one of two ways. Posterized rounds each channel to its nearest level. Dithered spreads the rounding error to the neighboring pixels (Floyd–Steinberg), so a patch of pixels averages to the original color.

Examples and queries

The examples are drawn from hats × huelight, barn × ramp, bikes × rafting. The queries are parrots × ramp and parrots × huelight, where "huelight" is the hue and lightness map.

Example walk

How each example pixel is chosen: at random from its example pair (the default below), or at the query pixel's own position, so the examples walk over their images in step with the query (E4).

True answer

The op applied to the pixel pair, rounded to the nearest grid level, which is the most likely answer under the random rounding the corpus uses.

Argmax and T = 1

The color the model gives the most probability, and one color drawn from the model's distribution over colors.

Full edit

The e_1 component removed from the state at every position and every slice, as in the D2.2 design.

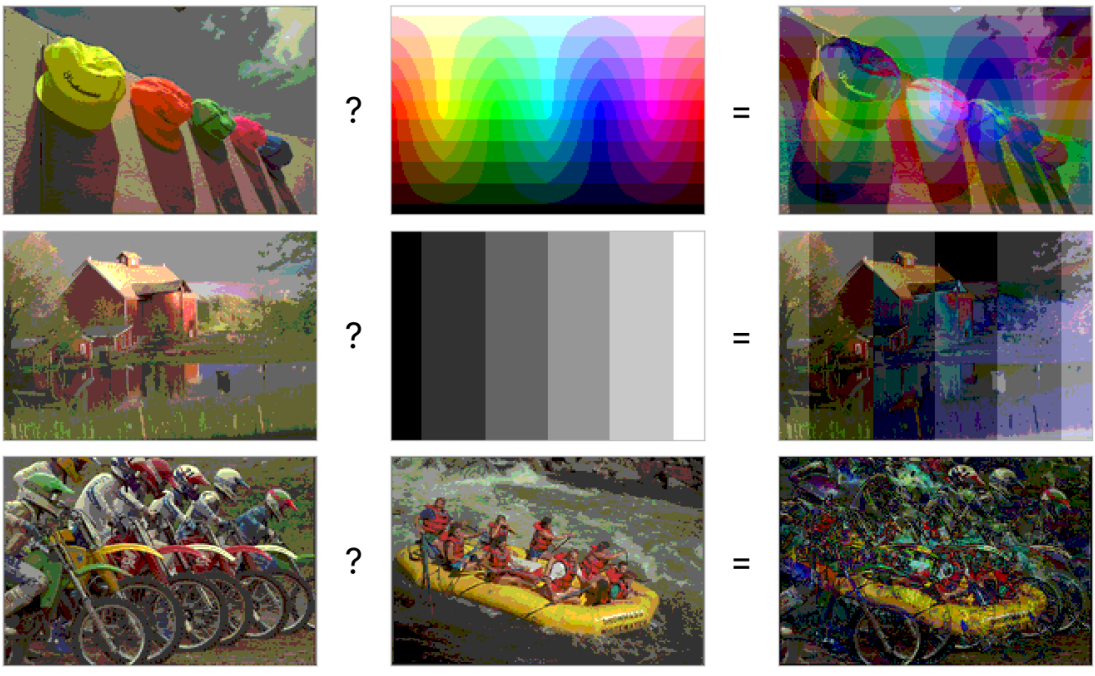
The pictures below are from seed 716, the median of the eight in how much of `difference` the edit takes out. The table in E5 covers all eight.

A context as pictures (E1)

What does one context look like when its operands° are pictures? Below are the three examples of `difference`, each with its true answer°, and the query, set off by a rule, with the model's answer in a bold frame. The model's answer is drawn at T = 1, and it is the only picture the model made.

Operands. Photos from the Kodak Lossless True Color suite, which Kodak released for unrestricted use, and two synthetic images…

True answer. The op applied to the pixel pair, rounded to the nearest grid level, which is the most likely answer under the random rounding the corpus uses.





The examples show the op on very different inputs: a hue and lightness map, a gray ramp, and two photos. The query picture is mostly clean. Under `difference` the parrots keep their colors where the ramp is dark and turn to their complementary colors where it is light, as the true op does, with a scatter of single pixels where the drawn examples left the op in doubt or the draw landed on a less likely color.

Taking out `difference` (E2)

What does the edit do to the picture? Each row is one query, with its operands°, then the true answer°, the unedited model, and the edited model, both models drawn at T = 1.

Operands. Photos from the Kodak Lossless True Color suite, which Kodak released for unrestricted use, and two synthetic images…

True answer. The op applied to the pixel pair, rounded to the nearest grid level, which is the most likely answer under the random rounding the corpus uses.

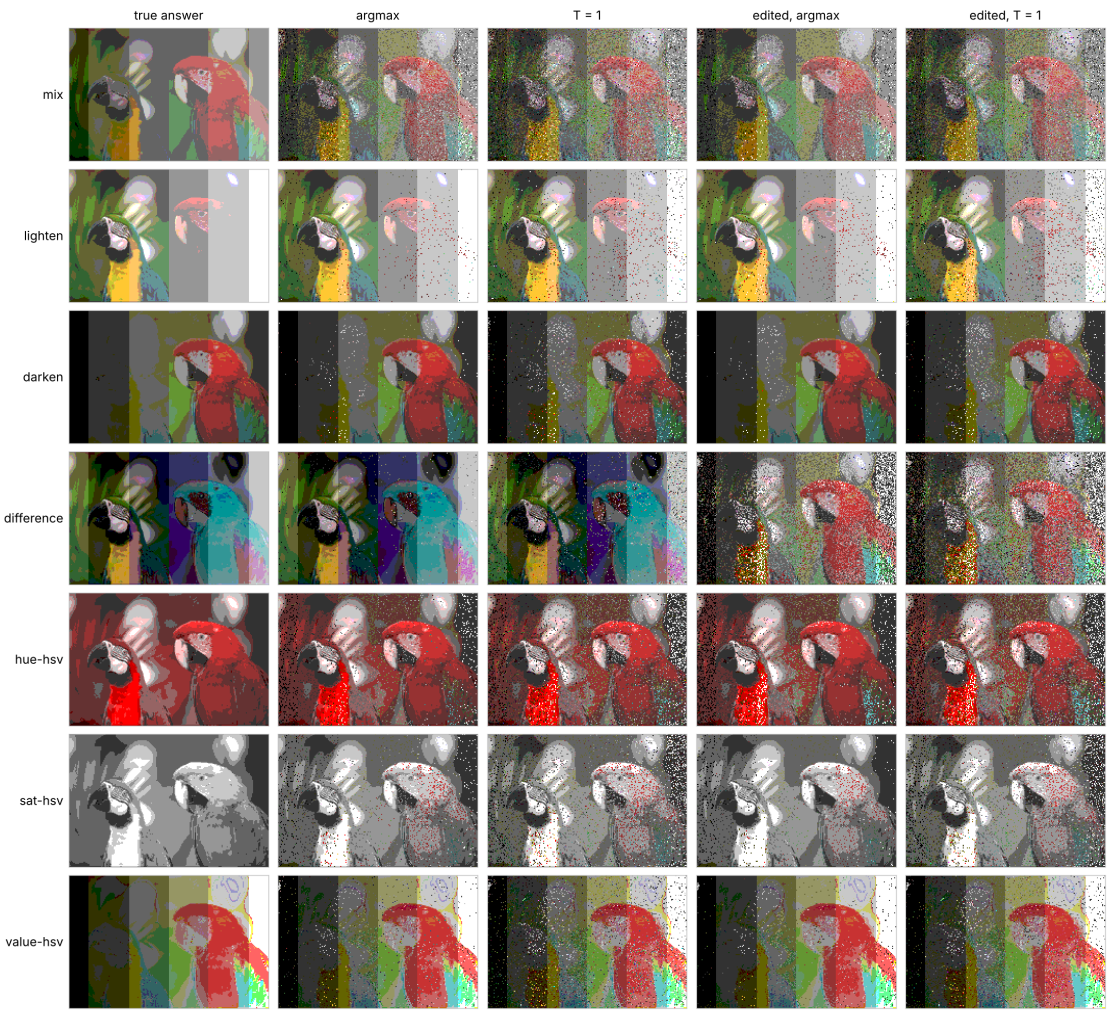


The unedited model paints `difference` on both queries. The edited model paints something else: on the gray ramp the parrots come back mostly in their own colors, and on the hue and lightness map the picture looks at first like `mix`. Looked at closely, it is a speckle of the other ops, pixel by pixel, which E3 shows side by side.

Every op, before and after the edit (E3)

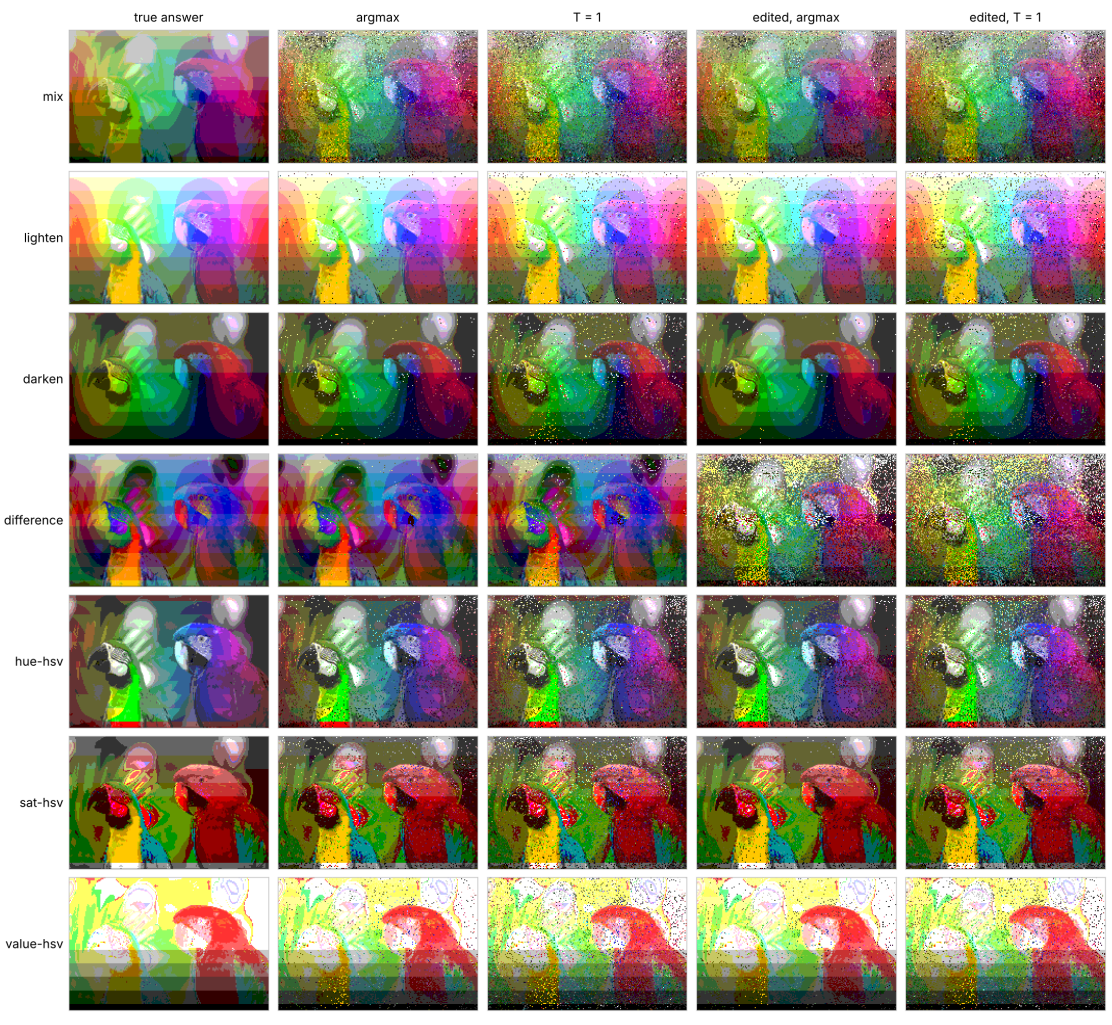
Does the picture carry the op for every op, and what does the edit do to each? Each row is one op, with the true answer°, then the unedited model, then the edited model. The query is parrots × ramp, posterized.

True answer. The op applied to the pixel pair, rounded to the nearest grid level, which is the most likely answer under the random rounding the corpus uses.



The unedited model paints each op recognizably. Most argmax pictures are close to the true answers, with a light speckle that the T = 1 pictures thicken. `mix` is speckled even at argmax: its answers round at random on most pairs in the corpus, so the model rightly spreads its probability over neighboring colors and the most likely one changes from pixel to pixel. Under the edit only the `difference` row changes. The argmax picture shows the op the edited model favors at each pixel, and the T = 1 picture shows the whole spread.

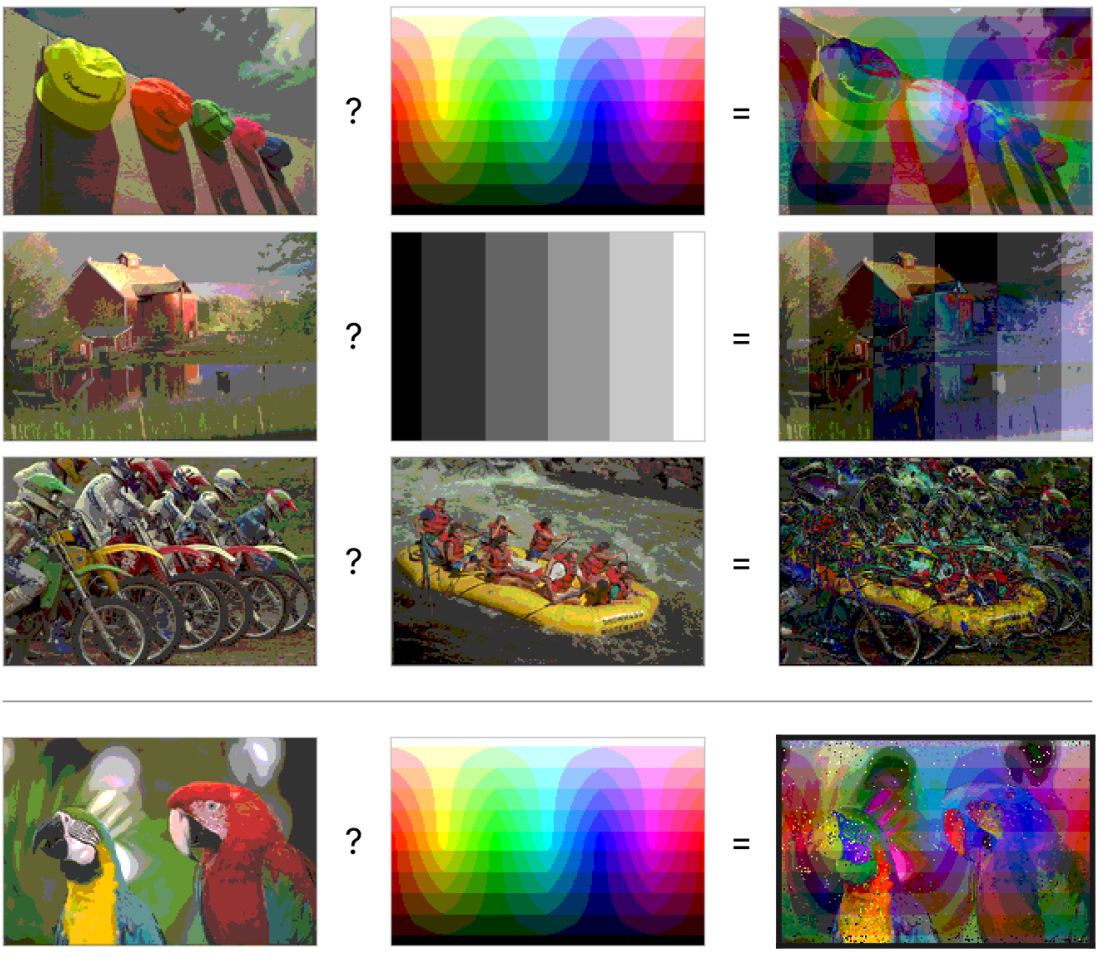
The same figure for the hue and lightness query:



Here the edited `difference` picture is closest to `mix`, with patches of `lighten`, `darken` and the HSV ops, which fits an edited model that falls back on the ops that best explain the examples at each pixel.

Examples in step with the query (E4)

So far each pixel drew its examples at random. A compositor works differently: it walks every image in step, so each pixel of the query sees the examples at its own position. What changes when the examples walk in step with the query? The same two figures, with the examples taken that way:



a b true answer unedited edited

Walking in step changes little overall: the probability on the true answer° is within a hundredth or two of the random walk for every op, edited or not (E5). What changes is where the stray pixels fall. With random examples they spread evenly over the picture. In step they gather into patches that follow the example images, presumably where the three examples at that position say little about the op, because several ops give them similar answers. The edited pictures show this most (on seed 716, a map of the probability on the true answer, averaged over 8 × 8 blocks, varies a third to a half more in step than at random, against about a sixth more for the unedited model), which fits an edited model that leans on the examples to choose among the remaining ops.

True answer. The op applied to the pixel pair, rounded to the nearest grid level, which is the most likely answer under the random rounding the corpus uses.

For a reader who knows compositors, the in-step walk is the familiar one, and its patches show something true about in-context inference: each answer depends on what its own examples show. For a reader new to both, the patches may look like part of the op, so the random walk keeps the picture closer to the op alone.

All eight runs (E5)

Do the other seeds paint the same pictures? The table gives the probability each run puts on the true answer°, averaged over the pixels of the parrots × ramp query (posterized), as the seed mean with the seed range in parentheses.

True answer. The op applied to the pixel pair, rounded to the nearest grid level, which is the most likely answer under the random rounding the corpus uses.

op	random, unedited	random, edited	in step, unedited	in step, edited
`mix`	0.41 (0.38 to 0.45)	0.41 (0.38 to 0.46)	0.42 (0.39 to 0.45)	0.41 (0.39 to 0.47)
`lighten`	0.88 (0.79 to 0.92)	0.85 (0.81 to 0.91)	0.88 (0.79 to 0.92)	0.85 (0.81 to 0.90)
`darken`	0.89 (0.85 to 0.92)	0.86 (0.79 to 0.90)	0.90 (0.85 to 0.93)	0.88 (0.79 to 0.92)
`difference`	0.88 (0.84 to 0.91)	0.08 (0.05 to 0.13)	0.89 (0.84 to 0.91)	0.09 (0.06 to 0.14)
`hue-hsv`	0.77 (0.69 to 0.84)	0.73 (0.65 to 0.86)	0.78 (0.71 to 0.86)	0.74 (0.68 to 0.88)
`sat-hsv`	0.76 (0.54 to 0.84)	0.74 (0.48 to 0.85)	0.77 (0.54 to 0.85)	0.75 (0.49 to 0.86)
`value-hsv`	0.64 (0.44 to 0.69)	0.64 (0.42 to 0.70)	0.66 (0.46 to 0.71)	0.65 (0.44 to 0.71)

With random examples the edit takes the `difference` picture from 0.88 to 0.08 on the seed mean and moves the other ops by a few hundredths at most, so the pictures of seed 716 are typical. These probabilities are not the D2.2 scores, which are taken on held-out corpus contexts with replacement op noise and weigh every answer the random rounding can give.

The panels as files

Every panel of seed 716 is also a PNG at its native size (192 × 128, one pixel per grid color), named in panels.json. A small page shows them as one context at a time, with a choice of op, second image, model, colors and example walk°, for embedding in a post.

Example walk. How each example pixel is chosen…

References

fixed-confirm

Fixed confirmation: `τ_ℓ fixed` kept the anchor on all eight fresh seeds, 712 to 719, and became the recipe of record.